Minimum Wage Pass-Through into Restaurant Prices: Empirical Design, Data, and Results

Empirical design, coding, and results: Arun Sinha

Background research and literature review: Amy Zheng, Aidan Teeling, Siddharth Chittapuram, Ayat Ashraf, Andrew Urner

June 17, 2026

1 Data Sources and Sample Selection

This project studies whether minimum-wage increases are passed through into consumer prices for food away from home, using a metro-month panel difference-in-differences design. The unit of observation is the metropolitan area by month. The outcome variable is the monthly change in the log Food Away From Home CPI index, which serves as a proxy for restaurant and prepared-food price inflation.

The final sample contains three metro areas: Los Angeles, New York, and Chicago. These cities are useful for this design because they experienced different minimum-wage policy paths over time while also having available monthly food-away-from-home CPI data. The cleaned panel covers February 2009 through February 2026, yielding 612 metro-month observations, with 204 monthly observations for each metro area.

The outcome data are built from metro-level Food Away From Home CPI series. Los Angeles is read from a BLS CPI workbook, while New York and Chicago are pulled from FRED CPI series. The project also uses a national all-items CPI series as a control for broader inflationary pressure that is not particular to restaurant labor costs.

The treatment data come from state and local minimum-wage policy files covering wage changes since 2009. State-level minimum-wage changes are read from the USDOL historical state table (2009-2024) and FRED annual state minimum wage release table (2025-2026), while city and county wage ordinances are read from the file containing data from the UC Berkeley Labor Center, Inventory of US City and County Minimum Wage Ordinances. The code explicitly incorporates the federal minimum wage, so that the effective binding wage in each metro reflects the maximum of the applicable federal, state, and mapped local wage rates.

Metro treatment exposure is constructed in two ways. The main treatment measure (`mw_core_local`) uses the maximum of the metro's core-state minimum wage and relevant local wage ordinances. For Los Angeles, mapped local jurisdictions include Los Angeles County, Los Angeles, Santa Monica, and Malibu. For Chicago, the local mapping includes Cook County. New York's treatment primarily reflects state-level exposure. A second treatment measure (`multistate_avg_rate`) is used as a robustness check for metros that span multiple states. In that version, the project averages the relevant state minimum-wage rates for each metro and then applies the federal minimum as a floor.

The main estimation sample begins in February 2009 rather than January 2009 since the dependent variable and treatment variables are first differences in logs. The first month is therefore lost when constructing monthly growth rates. Observations with missing values for the outcome, treatment, or inflation control are excluded from the estimation sample. The final treated-month indicator equals one in months when a metro experiences a positive increase in the constructed minimum-wage measure. Across the sample, there are 28 treated metro-months, meaning most of the identifying variation comes from a relatively small number of months wherein the policy changed.

2 Relationship to Existing Literature and Research Design Influences

This project builds on the minimum-wage price pass-through literature, especially those studies that use restaurants as a setting where wage-floor changes create direct labor-cost shocks. Katz and Krueger (1992) establish fast food as a useful low-wage industry for studying minimum-wage effects, while Aaronson, French, and MacDonald (2008) motivate the use of Food Away From Home CPI data to estimate whether restaurant prices rise after minimum-wage increases. this project adapts that approach using monthly metro-level Food Away From Home CPI series rather than confidential store-level BLS price data.

The empirical design also follows Aaronson, French, and MacDonald (2008) and MacDonald and Nilsson (2016) by treating the minimum wage as a continuous policy shock, measured as the log change in the applicable wage floor. This allows the main coefficient to be interpreted as a wage-price pass-through elasticity. The construction of an effective metro-level minimum wage, combining federal, state, and relevant local wage laws, is especially influenced by MacDonald and Nilsson's (2016) emphasis on state, city, and index minimum-wage variation.

The identification strategy is also informed by local minimum-wage studies such as Dube, Naidu, and Reich (2007) and Allegretto and Reich (2018), which use citywide policy changes and comparison groups to study restaurant outcomes. The dynamic lead-lag specification is motivated by MacDonald and Aaronson (2006), MacDonald and Nilsson (2016), and Fougere, Gautier, and Le Bihan (2010), who show that restaurant price adjustment may be immediate, staggered, or delayed because of price stickiness. Overall, this project is best understood as a transparent metro-month adaptation of the restaurant price pass-through literature, using publicly available CPI and minimum-wage data to estimate whether wage hikes are associated with higher Food Away From Home inflation.

3 Empirical Strategy: Identification Framework

The project uses a difference-in-differences framework to estimate the pass-through of minimum-wage increases into food-away-from-home prices. Rather than only comparing treated and untreated cities, the main specification uses the size of each wage increase as a continuous treatment. This makes the design an intensity difference-in-differences model: it asks whether months with larger percentage increases in the applicable minimum wage are associated with larger percentage changes in food-away-from-home prices, after controlling for common time patterns and persistent differences across metros.

Identification comes from variation in the timing and magnitude of minimum-wage increases across the three metros. The key comparison is within-metro price growth during months with minimum-wage increases versus months without increases, relative to contemporaneous price growth in other metros. Metro fixed effects absorb permanent differences across cities, such as baseline cost levels, persistent restaurant-market differences, and average price inflation differences. Month-of-year fixed effects absorb seasonality in restaurant prices. Year fixed effects absorb broad annual macroeconomic trends. The national CPI control further accounts for general inflationary pressure that may affect all prices, not just restaurant prices.

The identifying assumption is that, absent minimum-wage changes, treated and untreated metro-months would have followed similar food-away-from-home price trends after conditioning on fixed effects and controls. That is to say, the timing of minimum-wage hikes should not be systematically correlated with other unobserved metro-specific shocks to restaurant prices. For example, if a city raised its minimum wage at the same time that restaurant demand surged for unrelated reasons, the model could mistakenly attribute some of that demand-driven price growth to the wage hike.

The design also assumes that treatment timing is measured accurately and that the constructed wage measure captures the binding labor-cost shock faced by restaurants in each metro. This is particularly important because metropolitan areas do not always map perfectly onto state and local minimum-wage jurisdictions. The project addresses this by building both a core-state-plus-local treatment measure and a multistate-average robustness measure.

The dynamic specification provides an additional check on the timing assumption. By including a lead of the minimum-wage change, the model tests whether prices were already moving before the wage increase took effect. A large pre-treatment coefficient would weaken the causal interpretation because it would suggest anticipatory pricing, omitted local shocks, or non-parallel pre-trends. The lag term tests whether pass-through appears with delay rather than immediately in the policy month.

A major limitation is the small number of metro areas. Although the panel has many monthly observations, it has only three cross-sectional units and only 28 treated metro-months. Therefore, the estimated coefficients can be useful as exploratory evidence, but statistical inference should be interpreted cautiously, especially because clustering at the metro level leaves only three clusters.

4 Econometric Model

The main estimating equation is:

$$\begin{aligned}\Delta \log(FAFH_{m,t}) = \beta\Delta \log(MWm,t) + \gamma\Delta \log(CPI_t) \\ +\alpha_m + \lambda_{month(t)} + \delta_{year(t)} + \varepsilon_{m,t}\end{aligned} \tag{1}$$

where m indexes metropolitan areas and t indexes months.

The dependent variable, $\Delta \log(FAFH_{m,t})$, is the monthly log change in the metro-level Food Away From Home CPI index. This is interpreted as monthly restaurant-price inflation in metro m.

The main treatment variable, $\Delta \log(MWm,t)$, is the monthly log change in the constructed applicable minimum wage for the metro. In the baseline model, this is the `mw_core_local` measure, which combines the federal, core-state, and mapped local minimum-wage rates. The coefficient of interest is β. Because both the outcome and treatment are in log changes, β can be interpreted as a pass-through elasticity. For example, if $\beta = 0.10$, then a 10 percent minimum-wage increase would be associated with roughly a 1 percent increase in food-away-from-home prices.

The control variable, $\Delta \log(CPI_t)$, is the monthly log change in the national CPI. This assists in accounting for national inflation shocks that may influence restaurant prices across all metros.

The fixed effect α_m is a metro fixed effect. It controls for time-invariant differences across Los Angeles, New York, and Chicago. The term $\lambda_{month(t)}$ represents month-of-year fixed effects, which control for seasonal patterns in food-away-from-home inflation. The term $\delta_{year(t)}$ represents year fixed effects, which control for annual macroeconomic conditions and broad inflation regimes. Standard errors are clustered at the metro level.

This project also makes use of a binary treated-month model, which replaces the continuous wage-change variable with an indicator equal to one when the metro experiences a minimum-wage increase in that month:

$$\begin{aligned}\Delta \log(FAFH_{m,t}) = \tau Treat_{m,t} + \gamma\Delta \log(CPI_t) + \alpha_m + \lambda_{month(t)} \\ +\delta_{year(t)} + \varepsilon_{m,t}\end{aligned} \tag{2}$$

Here, τ measures whether food-away-from-home prices rise more in minimum-wage-hike months than in non-hike months, not considering the size of the wage increase.

The dynamic model includes lead, current, and lagged minimum-wage changes:

$$\begin{aligned}\Delta \log(FAFH_{m,t}) = \beta_{-1}\Delta \log(MW_{m,t+1}) + \beta_0 \Delta \log(MW_{m,t}) \\ +\beta_1 \Delta \log(MW_{m,t-1}) + \gamma \Delta \log(CPI_t) \\ +\alpha_m + \lambda_{month(t)} + \delta_{year(t)} + \varepsilon_{m,t}\end{aligned} \tag{3}$$

In this model, the lead term helps assess whether prices were moving before the policy change, the current term captures immediate pass-through, and the lag term captures delayed pass-through. The sum of the lead, current, and lag coefficients provides a rough cumulative estimate, though the lead should be interpreted mainly as a diagnostic for pre-trends or anticipation.

Finally, the project robustness checks using the multistate-average minimum-wage measure and stricter time fixed-effect structure with metro fixed effects and year-month fixed effects:

$$\Delta \log(FAFH_{m,t}) = \beta \Delta \log(MW_{m,t+1}) + \alpha_m + \theta_t + \varepsilon_{m,t} \tag{4}$$

where θ_t is a year-month fixed effect. This specification absorbs shocks common to all metros in the same calendar month, making identification come only from differences in wage changes across metros within the same month. Because this is a more demanding specification, it is useful as a robustness check for whether the eliminated pass-through survives after controlling for national month-specific shocks.

5 Results and Interpretation

The results provide weak and statistically inconclusive evidence that minimum-wage increases were passed through into Food Away From Home CPI inflation in this three-metro panel. In the main intensity differences-in-differences model, the coefficient on the log change in the core local minimum wage is positive but very small and statistically insignificant. The binary treated-month model reaches a similar conclusion: months with minimum-wage increases do not show a clear or statistically meaningful price jump relative to untreated months.

The robustness checks generally also point to the same result. the equal-weight multistate treatment specification produces a positive but statistically weak estimate, suggesting that the main finding is not only driven by one method of mapping minimum-wage exposure to metropolitan areas. The stricter model with metro fixed effects and year-month fixed effects also remains positive but insignificant. Meanwhile, the dynamic lead-current-lag model does not show a clear pattern of positive pass-through; the dynamic model yields a contemporaneous coefficient of nearly zero, while the lead and lag terms are imprecisely estimated.

Overall, the study only considered three metropolitan areas and 28 treated metro-months, which limits statistical power and makes inference dubious. The safest conclusion is that this public CPI-based metro panel does not reproduce the larger positive pass-through estimates found in parts of the prior literature. Instead, the results suggest that any minimum-wage effect on Food Away From Home prices is difficult to detect in this sample.